

$$T(TA \otimes TB) \begin{array}{c} \xrightarrow{\mu_{A \otimes B} \circ T(\phi_{A,B})} \\ \xrightarrow{T(a_A \otimes a_B)} \end{array} T(A \otimes B) \xrightarrow{q} A \otimes^{\phi} B.$$